

# NIRIKSHAK PHYSICAL VALIDATION DATASET

Smart India Hackathon 2026 · Ministry of Consumer Affairs · Physical Evidence Dossier

## Item 2 — General Wellness: Himalaya Brahmi Mind Wellness (60 Tablets)

|                       |                                                         |                         |                                                            |
|-----------------------|---------------------------------------------------------|-------------------------|------------------------------------------------------------|
| Product / Brand:      | Himalaya (Since 1930) Pure Herbs · Brahmi Mind Wellness | Commodity / Category:   | Ayurvedic Medicine (Consumer Health / Wellness)            |
| Net Quantity:         | 60 Tablets (Count declaration)                          | Unit Sale Price (USP):  | Rs. 4.33/TAB. (Explicit statutory print)                   |
| Maximum Retail Price: | Rs. 260.00 (Incl. of all taxes)                         | Batch / Lot Number:     | 372600253 (Mfg plant code: 37)                             |
| Mfg. Date / Expiry:   | 05/2026 – 04/2029 (36 months shelf life)                | Barcode (EAN-13):       | 8901138834265 · Code: IND-5061447                          |
| Manufacturer:         | Himalaya Wellness Company (Peenya, Bengaluru 560 058)   | Consumer Care Helpline: | Toll-Free: 1-800-208-1930   contactus@himalayawellness.com |
| CPCB Registration:    | BO-22-000-07-AADFT3025B-22                              | Active Ingredient:      | Brahmi (Bacopa monnieri) Wh. Pl. Extract 250 mg/Tab        |

## Physical Measurement & Spatial Calibration Ground Truth

| Metric / Parameter       | Reading 1       | Reading 2       | Calibration Reference      | Instrument & Least Count          |
|--------------------------|-----------------|-----------------|----------------------------|-----------------------------------|
| Carton Width (Front)     | 65.20 mm        | 65.10 mm        | In-plane SBI Card Scale    | Digital Caliper (0.01 mm LC)      |
| Carton Height (Vertical) | 88.20 mm        | 88.10 mm        | In-plane SBI Card Scale    | Digital Caliper (0.01 mm LC)      |
| Carton Depth (Lateral)   | 60.10 mm        | 60.00 mm        | In-plane SBI Card Scale    | Digital Caliper (0.01 mm LC)      |
| Carton Volume / PDP Area | 343.2 cm³       | 57.2 cm² (PDP)  | Calculated (W x H x D)     | LM Rules Rule 9 Table-I Tier      |
| SBI RuPay Reference Card | 85.60 mm (W)    | 53.98 mm (H)    | ISO/IEC 7810 ID-1 Standard | 18.69 px/mm native resolution     |
| MRP / USP Numeral Height | 1.47 mm (27 px) | 1.85 mm (34 px) | Calibrated against Card    | Pixel Profilometry / Optical Grid |

## Legal Metrology (Packaged Commodities) Rules, 2011 Compliance Audit

| Statutory Rule | Mandatory Declaration                            | Verbatim Package Ground Truth                                                                                     | Finding / Status                            |
|----------------|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| Rule 6(1)(a)   | Manufacturer / Packer Name & Complete Address    | Himalaya Wellness Company; At: (37) No. 473/C, 13th Cross, 4th Phase, Peenya Industrial Estate, Bengaluru 560 058 | COMPLIANT (Batch prefix 37 maps to address) |
| Rule 6(1)(b)   | Generic / Common Name of Commodity               | 'Brahmi MIND WELLNESS' / 'BACOPA' / 'Ayurvedic Medicine'                                                          | COMPLIANT (Prominent on all panels)         |
| Rule 6(1)(c)   | Net Quantity in Standard Metric / Count Unit     | 'Net Quantity: 60 Tablets' declared on Left & Front panels                                                        | COMPLIANT (Standard count specification)    |
| Rule 6(1)(d)   | Month & Year of Manufacture & Batch No.          | '05/2026-04/2029'; Batch No. '372600253'                                                                          | COMPLIANT (Clear 36-month period)           |
| Rule 6(1)(da)  | Unit Sale Price (USP) (Mandatory > 1 Unit)       | 'RS. 4.33/TAB.' explicitly printed next to MRP                                                                    | COMPLIANT (2022 LM Amendment)               |
| Rule 6(1)(e)   | Maximum Retail Price (MRP inclusive of taxes)    | 'RS. 260. 00' with 'MRP Rs. Incl. of all taxes and USP'                                                           | COMPLIANT (Explicit tax inclusion text)     |
| Rule 6(1)(f)   | Best Before / Expiry Date (Consumer Health)      | '04/2029' printed in conjunction with mfg date                                                                    | COMPLIANT (Ayurvedic shelf-life mandated)   |
| Rule 6(1)(g)   | Consumer Care Helpline, Email & Contact Person   | Manager - Customer Care; 1-800-208-1930; contactus@himalayawellness.com                                           | COMPLIANT (Toll-free + Email + Person)      |
| Rule 6(1)(h)   | Country of Origin Declaration                    | Indigenous Indian manufacture confirmed by Karnataka facility address                                             | COMPLIANT (Made in India)                   |
| Rule 9 Table-I | Minimum Numeral Height (PDP 50–200 cm²: >= 2 mm) | Dot-matrix print = 1.47–1.85 mm; Offset text = 1.53 mm                                                            | NOTEWORTHY (Sub-2mm inkjet print)           |
| CPCB EPR       | Plastic Waste / Central Pollution Control Board  | CPCB Number: BO-22-000-07-AADFT3025B-22                                                                           | COMPLIANT (Statutory Registration)          |

IMAGE #01: front\_01.jpg (orig: IMG\_20260911\_153123.jpg)

Standard Guide #1 · Principal Display Panel (PDP)



Target Packaging Face: Front Principal Display Panel (PDP)

Camera Angle & View: Orthographic perpendicular straight-on (90° optical axis)

Lighting / Illumination: Natural diffused ambient daylight; zero specular glare

Camera EXIF: ISO 641 · 1/25s (0.040s) · f/1.79 · 4.74mm · Flash Off (16) · 4608x3456 native

**Visible Verbatim Declarations:**

- Brand Heritage: 'Himalaya SINCE 1930 Pure Herbs'
- Product Name: 'Brahmi MIND WELLNESS'
- Active Botanical: 'BACOPA'
- Functional Claim: 'Improves alertness'
- Net Quantity: '60 Tablets' (in orange badge on bottom-right)

*Nirikshak Benchmark Significance: Benchmark baseline capture for PDP detection, brand logo segmentation, and claim extraction. High contrast dark green text on white substrate. SBI RuPay card standing vertically to right provides uncompromised spatial ground truth.*

Native Dim: 4608 x 3456 px (16 MP)

Spatial Scale: 18.69 px/mm (Card H: 1600 px / 85.60 mm, W: 858 px / 53.98 mm)

Physical Box: Width: 65.2 mm (1051 px) · Height: 88.2 mm (1643 px)

IMAGE #02: right\_01.jpg (orig: IMG\_20260911\_153134.jpg)

Standard Guide #4 · Lateral Barcode Panel



Target Packaging Face: Right Lateral Side Panel (Barcode & Corporate Vision)

Camera Angle & View: Orthographic perpendicular straight-on (90° optical axis)

Lighting / Illumination: Natural diffused ambient daylight; clean illumination

Camera EXIF: ISO 388 · 1/25s (0.040s) · f/1.79 · 4.74mm · Flash Off (16) · 4608x3456 native

**Visible Verbatim Declarations:**

- Barcode (EAN-13): '8901138834265' (GS1 India country code 890)
- Internal Code: 'IND-5061447'
- Corporate Vision: Complete Himalaya 3-paragraph science & wellness charter
- Storage / Advisory: 'Keep tightly closed... Store in a dry place away from heat & sunlight. Do not refrigerate. Keep out of reach of children.'
- Intellectual Property: '© Regd. Trademark · Trademark Owner: Himalaya Global Holdings Ltd. · Licensed User: Himalaya Wellness Company'

*Nirikshak Benchmark Significance: Essential for 1D Barcode localization and 100% GS1 decodability validation. Tests OCR on dense multi-line fine-print legal and storage texts.*

Native Dim: 4608 x 3456 px (16 MP)

Spatial Scale: 18.45 px/mm (Card H: 1579 px / 85.60 mm)

Physical Box: Depth: 60.1 mm (1108 px) · Height: 88.1 mm (1625 px)

IMAGE #03: back\_01.jpg (orig: IMG\_20260911\_153142.jpg)

## Standard Guide #2 · Opposite / Back Panel



Native Dim: 4608 x 3456 px (16 MP)  
Spatial Scale: 18.52 px/mm (Card H: 1585 px / 85.60 mm)  
Physical Box: Width: 65.1 mm (1050 px) · Height: 88.1 mm (1631 px)

Target Packaging Face: Back Opposite Panel (Formulation, Dosage & Environmental EPR)

Camera Angle & View: Orthographic perpendicular straight-on (90° optical axis)

Lighting / Illumination: Natural diffused ambient daylight; balanced illumination

Camera EXIF: ISO 360 · 1/25s (0.040s) · f/1.79 · 4.74mm · Flash Off (16) · 4608x3456 native

**Visible Verbatim Declarations:**

- Product Identity: 'Himalaya SINCE 1930 Pure Herbs Brahmi MIND WELLNESS'
- Formulation Composition: 'EACH TABLET CONTAINS: EXTRACT: Brahmi (Bacopa monnieri) Wh. Pl. 250 mg'
- Pharmacopoeial Reference: 'Ref.: # Bhava Prakasha'
- Environmental Registration: 'CPCB Number: BO-22-000-07-AADFT3025B-22'
- Recommended Dosage: 'DOSAGE: 1-2 tablets twice daily, or as directed by your doctor.'

*Nirikshak Benchmark Significance: Critical back panel evidence for regulatory drug/ayurvedic formulation extraction, botanical Latin name verification (Bacopa monnieri), and statutory CPCB registration number OCR.*

IMAGE #04: left\_01.jpg (orig: IMG\_20260911\_153152.jpg)

## Standard Guide #3 · Primary Legal Metrology Declarations



Native Dim: 4608 x 3456 px (16 MP)  
Spatial Scale: 18.33 px/mm (Card H: 1569 px / 85.60 mm)  
Physical Box: Depth: 60.1 mm (1102 px) · Height: 88.0 mm (1613 px)

Target Packaging Face: Left Lateral Side Panel (Statutory Compliance Hub)

Camera Angle & View: Orthographic perpendicular straight-on (90° optical axis)

Lighting / Illumination: Evenly diffused daylight; pin-sharp legibility

Camera EXIF: ISO 439 · 1/33s (0.030s) · f/1.79 · 4.74mm · Flash Off (16) · 4608x3456 native

**Visible Verbatim Declarations:**

- Net Quantity: 'Net Quantity: 60 Tablets'
- Batch Number: '372600253' (Dot-matrix inkjet)
- Mfg & Expiry Dates: '05/2026-04/2029' (Dot-matrix inkjet)
- Maximum Retail Price: 'RS. 260. 00' (Dot-matrix inkjet)
- Unit Sale Price: 'RS. 4.33/TAB.' (Explicit Rule 6(1)(da) print)
- Manufacturer Multi-Site Address: '4 licensed manufacturing locations (Peenya [37], Pudukcherry [72], Nelamangala [106], Tumakuru [111])'
- Consumer Care: Manager - Customer Care, Makali Bengaluru 562 162 | Toll-free: 1-800-208-1930 | Email: contactus@himalayawellness.com

*Nirikshak Benchmark Significance: THE PRIMARY LEGAL METROLOGY DECLARATION PANEL. Tests OCR on mixed print technologies: offset typography (Net Qty, address, helpline) combined with industrial dot-matrix/inkjet (Batch, Dates, MRP, USP). Benchmark target for Rule 6 compliance verification.*

IMAGE #05: top\_01.jpg (orig: IMG\_20260911\_153209.jpg)Guide Extension · Top Closure Flap



Target Packaging Face: Top Flap Closure Panel  
Camera Angle & View: Orthographic perpendicular straight-on (90° optical axis)  
Lighting / Illumination: Natural diffused ambient daylight; flat field  
Camera EXIF: ISO 405 · 1/25s (0.040s) · f/1.79 · 4.74mm · Flash Off (16) · 4608x3456 native

**Visible Verbatim Declarations:**

- Brand Logo & Header: 'Himalaya SINCE 1930'
- Commodity Name: 'Brahmi MIND WELLNESS'

*Nirikshak Benchmark Significance: Verifies top closure integrity and confirms that no dispersed mandatory declarations are concealed on secondary packaging flaps. Top dimensions establish carton cross-sectional profile (65 mm x 60 mm).*

Native Dim: 4608 x 3456 px (16 MP)  
Spatial Scale: 18.61 px/mm (Card H: 1593 px / 85.60 mm)  
Physical Box: Width: 65.0 mm (1210 px) · Depth: 60.0 mm (1117 px)

IMAGE #06: bottom\_01.jpg (orig: IMG\_20260911\_153215.jpg)Guide Extension · Bottom Base Panel



Target Packaging Face: Bottom Base Tuck-in Flap  
Camera Angle & View: Orthographic perpendicular straight-on (90° optical axis)  
Lighting / Illumination: Natural ambient lighting; high exposure sensitivity (ISO 888)  
Camera EXIF: ISO 888 · 1/25s (0.040s) · f/1.79 · 4.74mm · Flash Off (16) · 4608x3456 native

**Visible Verbatim Declarations:**

- Base Surface: Clean unprinted solid white folding carton base with orange accent stripe.
- Mandatory Declarations: NONE (Base panel verified free of hidden markings).

*Nirikshak Benchmark Significance: Guarantees 100% full 6-sided 3D package enclosure verification under Nirikshak rules. Negative-evidence baseline ensuring no hidden or misplaced statutory declarations exist.*

Native Dim: 4608 x 3456 px (16 MP)  
Spatial Scale: 18.58 px/mm (Card H: 1590 px / 85.60 mm)  
Physical Box: Width: 65.1 mm (1209 px) · Depth: 60.0 mm (1115 px)



## IMAGE #07: front\_angle\_01.jpg (orig: IMG\_20260911\_153312.jpg)

## Standard Guide #5 · Moderate Real Angle



Native Dim: 4608 x 3456 px (16 MP)  
Spatial Scale: Perspective keystone (~16.2 to 19.8 px/mm gradient)  
Physical Box: Perspective 3D Projection (Front + Left + Top bevel)

Target Packaging Face: Dual-Face Perspective: Front PDP + Left Legal Metrology Panel

Camera Angle & View: Moderate perspective oblique angle (~25°–30° elevation, ~35° yaw)

Lighting / Illumination: Diffused directional daylight with specular edge highlights

Camera EXIF: ISO 295 · 1/33s (0.030s) · f/1.79 · 4.74mm · Flash Off (16) · 4608x3456 native

**Visible Verbatim Declarations:**

- Front Panel: 'Himalaya SINCE 1930 Pure Herbs Brahmi MIND WELLNESS BACOPA'
- Left Panel: 'Net Quantity: 60 Tablets', '372600253', '05/2026-04/2029', 'RS. 260. 00', 'RS. 4.33/TAB.'
- Top Flap: Visible at receding angle ('Brahmi MIND WELLNESS')

*Nirikshak Benchmark Significance: Tests AI perspective rectification, multi-panel bounding polygon segmentation, and keystone-distorted text OCR. Simultaneously captures brand identity and statutory pricing.*

## IMAGE #08: side\_angle\_01.jpg (orig: IMG\_20260911\_153315.jpg)

## Custom Addition · Oblique Declaration View



Native Dim: 4608 x 3456 px (16 MP)  
Spatial Scale: Perspective keystone (~16.5 to 19.5 px/mm gradient)  
Physical Box: Perspective 3D Projection (Left declaration wall emphasis)

Target Packaging Face: Oblique Perspective Focused on Legal Metrology Panel  
Camera Angle & View: Moderate oblique angle (~30° elevation, focused on Left Panel)

Lighting / Illumination: Balanced ambient lighting with suppressed specular highlights

Camera EXIF: ISO 295 · 1/33s (0.030s) · f/1.79 · 4.74mm · Flash Off (16) · 4608x3456 native

**Visible Verbatim Declarations:**

- Mandatory Hub: Net Quantity, Batch No., Mfg/Exp Dates, MRP, USP
- Address Grid: Peenya (37), Puducherry (72), Nelamangala (106), Tumakuru (111)
- Helpline: 1-800-208-1930 & contactus@himalayawellness.com

*Nirikshak Benchmark Significance: Custom angle capture tailored for angle-dependent OCR reliability. Captures dot-matrix characters under natural surface micro-texture without glare blowout.*

IMAGE #09: back\_angle\_01.jpg (orig: IMG\_20260911\_153400.jpg)Standard Guide #6 · Moderate Real Angle Back



Target Packaging Face: Dual-Face Perspective: Back Panel + Right Barcode Panel  
Camera Angle & View: Moderate perspective oblique angle (~30°–45° elevation)  
Lighting / Illumination: Diffused directional daylight with smooth corner shading  
Camera EXIF: ISO 439 · 1/33s (0.030s) · f/1.79 · 4.74mm · Flash Off (16) · 4608x3456 native

**Visible Verbatim Declarations:**

- Back Panel: 'Brahmi (Bacopa monnieri) 250 mg', 'CPCB Number: BO-22-000-07-AADFT3025B-22'
- Right Panel: EAN-13 Barcode '8901138834265', Code 'IND-5061447', Himalaya Vision charter

*Nirikshak Benchmark Significance: Benchmark test for simultaneous barcode detection on an oblique planar surface combined with Latin botanical nomenclature and CPCB registration OCR.*

Native Dim: 4608 x 3456 px (16 MP)  
Spatial Scale: Perspective keystoned (~16.0 to 19.2 px/mm gradient)  
Physical Box: Perspective 3D Projection (Back + Right lateral wall)

IMAGE #10: glare\_01.jpg (orig: IMG\_20260911\_153416.jpg)Standard Guide #7 · Direct Flash Specular Glare



Target Packaging Face: Left Legal Metrology Panel under Direct Flash  
Camera Angle & View: Orthogonal vertical top-down with direct coaxial strobe  
Lighting / Illumination: High-intensity camera flash strobe; 1/100s sync; dark perimeter  
Camera EXIF: ISO 92 · 1/100s (0.010s) · f/1.79 · 4.74mm · Flash Fired (1) · 4608x3456 native

**Visible Verbatim Declarations:**

- Net Quantity: 'Net Quantity: 60 Tablets'
- Batch No. & Dates: '372600253' · '05/2026-04/2029'
- MRP & USP: 'RS. 260. 00' · 'RS. 4.33/TAB.' (Subject to specular hotspot)
- Manufacturer Panel: Complete Peenya & multi-site licensing text

*Nirikshak Benchmark Significance: RIGOROUS STRESS TEST: Camera flash fired directly at glossy/matte cardboard coating, generating realistic specular glare hotspot across the Net Qty, MRP, and USP fields. Tests AI OCR resilience against flash wash-out, localized saturation, and high-frequency noise.*

Native Dim: 4608 x 3456 px (16 MP)  
Spatial Scale: 18.30 px/mm (High-speed flash sync)  
Physical Box: Depth: 60.1 mm · Height: 88.0 mm (high-contrast boundary)

IMAGE #11: glare\_front\_01.jpg (orig: IMG\_20260911\_153424.jpg)Custom Addition · Front Flash Glare Hotspot



Target Packaging Face: Front Principal Display Panel under Direct Flash  
Camera Angle & View: Orthogonal vertical top-down with direct coaxial strobe  
Lighting / Illumination: High-intensity strobe; ultra-fast shutter (1/205s); ISO 53; dark background  
Camera EXIF: ISO 53 · 1/205s (0.005s) · f/1.79 · 4.74mm · Flash Fired (1) · 4608x3456 native

**Visible Verbatim Declarations:**

- Front Artwork & Text: 'Himalaya SINCE 1930 Pure Herbs'
- Product Title: 'Brahmi MIND WELLNESS BACOPA'
- Claim & Count: 'Improves alertness · 60 Tablets'
- Glare Phenomenon: Intense circular specular reflection over Brahmi flower

*Nirikshak Benchmark Significance: Evaluates visual model feature extraction on front graphics and typography under extreme strobe lighting. Low ISO (53) and 1/205s shutter create black falloff.*

Native Dim: 4608 x 3456 px (16 MP)  
Spatial Scale: 18.65 px/mm (Card in dark field)  
Physical Box: Width: 65.2 mm · Height: 88.2 mm

IMAGE #12: glare\_barcode\_01.jpg (orig: IMG\_20260911\_153430.jpg)Custom Addition · Barcode Flash Specular Reflection



Target Packaging Face: Right Lateral Barcode Panel under Direct Flash  
Camera Angle & View: Orthogonal vertical top-down with direct coaxial strobe  
Lighting / Illumination: High-intensity flash strobe; 1/355s ultra-fast shutter; ISO 53  
Camera EXIF: ISO 53 · 1/355s (0.003s) · f/1.79 · 4.74mm · Flash Fired (1) · 4608x3456 native

**Visible Verbatim Declarations:**

- Barcode EAN-13: '8901138834265' under direct flash illumination
- SKU Code: 'IND-5061447'
- Himalaya Vision: Complete vision and philosophy charter
- Trademark Owner: 'Himalaya Global Holdings Ltd.'

*Nirikshak Benchmark Significance: Extreme stress test for 1D barcode scanners and AI decoders. Strobe flash creates intense white clipping across black bars, testing edge detection and binarization thresholding.*

Native Dim: 4608 x 3456 px (16 MP)  
Spatial Scale: 18.42 px/mm (High-speed flash sync)  
Physical Box: Depth: 60.1 mm · Height: 88.1 mm

# NIRIKSHAK DATASET ARCHITECTURE & VALIDATION METHODOLOGY

Smart India Hackathon 2026 · AI Vision Benchmark Standards · Quality Assurance Protocol

## 1. Dataset Taxonomy & Standard Guideline Alignment

This dataset strictly complies with the official guidelines stipulated in Nirikshak\_Physical\_Validation\_Dataset\_Guide\_SIH2026.pdf. The collection encompasses 12 high-resolution native captures (15.93 Megapixels per image, 4608 x 3456 px, uncompressed JPEG) capturing 100% of the 3D surface geometry of the physical packaged commodity across all 6 orthographic faces (Front, Back, Left, Right, Top, Bottom) supplemented by multi-angle perspective projections and realistic physical stress challenges. Per Section 2 of the Guide, standard nomenclature is preserved (front\_01.jpg, back\_01.jpg, left\_01.jpg, right\_01.jpg, front\_angle\_01.jpg, back\_angle\_01.jpg, glare\_01.jpg), while logical extensions (top\_01.jpg, bottom\_01.jpg) and custom angle additions (side\_angle\_01.jpg, glare\_front\_01.jpg, glare\_barcode\_01.jpg) have been integrated to guarantee zero blind spots.

## 2. Physical Calibration Protocol (ISO/IEC 7810 ID-1 Standard)

To satisfy the rigorous requirements of Section 4 ('Calibration / Reference Photo') and Section 6 ('Font / Declaration Height Measurements'), a standardized physical reference target—a State Bank of India (SBI) Global RuPay Debit Card—was co-placed within the field of view in the exact coplanar geometric alignment as the product face. Under international standard ISO/IEC 7810 ID-1, the physical dimensions are precisely 85.60 mm (width) x 53.98 mm (height) x 0.76 mm (thickness) with corner radii of 3.18 mm. At a native resolution of 4608 x 3456 px, this yields a calibrated spatial scale of 18.33 to 18.69 pixels per millimetre. This sub-millimetre resolution allows mathematical verification of physical character heights without digital extrapolation or synthetic assumption.

## 3. Five Core AI Stress-Testing Dimensions

| Benchmark Dimension       | Physical Capture Condition                                      | Associated Test Images                       | Target Evaluation Metric                                       |
|---------------------------|-----------------------------------------------------------------|----------------------------------------------|----------------------------------------------------------------|
| Orthographic OCR          | Perpendicular 90° optical axis, natural diffused daylight       | front_01, left_01, back_01, right_01         | Character Error Rate (CER) < 1%, Word Error Rate (WER) < 2%    |
| Perspective Invariance    | Oblique 25°–45° tilt, simultaneous multi-face keystoning        | front_angle_01, back_angle_01, side_angle_01 | Homography rectification & 3D bounding box IoU > 0.85          |
| Specular Glare Robustness | Direct coaxial camera flash strobe at fast sync (1/100s–1/355s) | glare_01, glare_front_01, glare_barcode_01   | OCR resilience under localized specular saturation & blooming  |
| 1D Barcode Decodability   | EAN-13 thermal/offset barcode under flat & flash lighting       | right_01, glare_barcode_01, back_angle_01    | ISO/IEC 15416 print quality grade decoding at > 99.5% accuracy |
| Micro-Print Metrology     | Industrial dot-matrix / inkjet character profiling (1.47 mm)    | left_01, glare_01                            | Rule 9 Table-I height threshold verification (MAE < 0.1 mm)    |

## 4. Submission Sign-Off & Integrity Declaration

*DATASET COMPLIANCE DECLARATION: All 12 photographic assets incorporated in this dossier represent authentic, unaltered raw optical evidence captured from genuine commercial physical packaged stock. No synthetic upscaling, sharpening, filtering, generative AI inpainting, or WhatsApp compression has been applied to any image. All legal metrology declarations transcribed herein correspond verbatim to the physical artifact under test. The dossier and dataset are fully validated for integration into the Nirikshak SIH 2026 AI Validation Suite.*

|                                                                      |                                                                          |                                                 |                              |
|----------------------------------------------------------------------|--------------------------------------------------------------------------|-------------------------------------------------|------------------------------|
| Dossier Compiled By:<br>Lead Data Metrologist / AI Research Engineer | Verification Authority:<br>Nirikshak Physical Validation Team · SIH 2026 | Audit Date & Timestamp:<br>2026-09-11 19:25 IST | Status:<br>VERIFIED & FROZEN |
|----------------------------------------------------------------------|--------------------------------------------------------------------------|-------------------------------------------------|------------------------------|